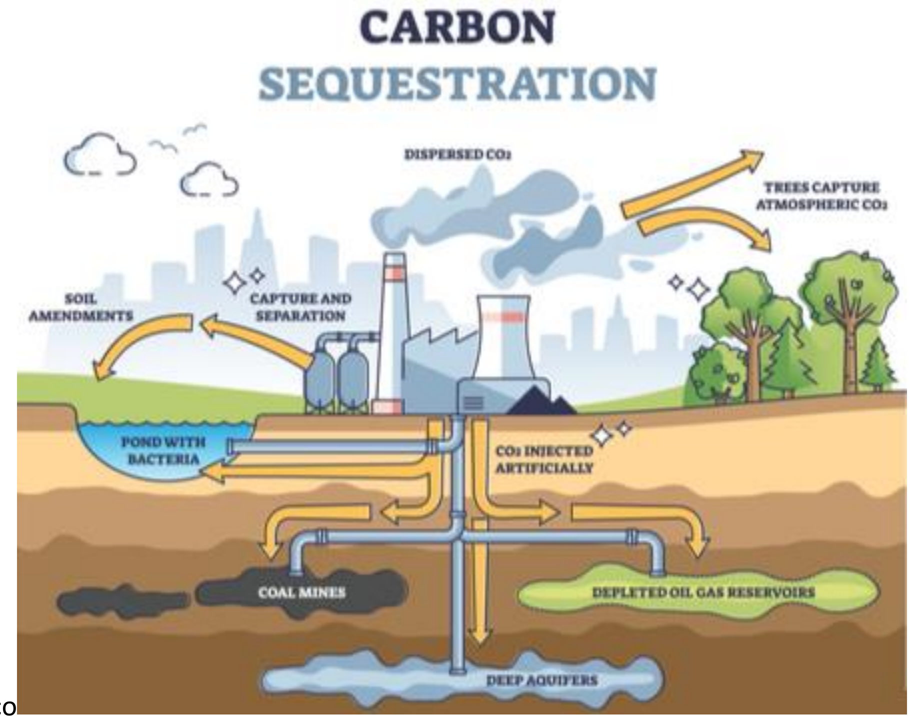
Carbon Sequestration

Carbon sequestration is the long-term storage of carbon dioxide or other forms of carbon with the aim of mitigating global warming and climate change. The natural carbon cycle is responsible for the exchange of carbon among the biosphere, pedosphere (soil layer), geosphere (rocks), hydrosphere and the atmosphere. Human activities have disrupted the carbon cycle since the past few centuries by the modification of land use and by excessive burning of fossil fuels. As a result, CO_2 has increased by over 52% higher than pre-industrial levels as of the year 2020. This increased CO_2 has disbalanced the carbon cycle, causing catastrophic changes in the climate, which we discussed in the section 1 of this module. Therefore, it is essential for us to restore this balance by removing excess CO_2 from the atmosphere by artificial methods and depositing them in a reservoir. This can be achieved using biological, physical and chemical methods.



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Natural Carbon Sinks

Carbon sink is a natural reservoir that absorbs and stores CO_2 from the atmosphere in the form of various carbon-based compounds. This is known as the natural process of carbon sequestration. In order to understand these completely, it is necessary for us to understand the carbon cycle, which is the mechanism of transfer of carbon between the atmosphere, lithosphere and hydrosphere. There are two major types of carbon sinks, namely terrestrial and oceanic sinks.

- Terrestrial sinks include soil, grasslands, trees, plants and any organic matter that act as both long-term and short-term sinks. Soil contains more carbon than all terrestrial vegetation and atmosphere combined.
- Ocean sinks: are the world's primary long-term carbon sink, absorbing more than 25% of CO_2 emitted by humans. Plankton and aquatic life in the oceans absorb CO_2 via photosynthesis; eventually die eventually, sink to the bottom, carrying the carbon deposits with them. When they decompose, the carbon-based compounds are transformed into other forms, utilized by the other aquatic species at the bottom of the ocean.

Terrestrial Biosequestration

Biosequestration can be utilized for the capture and storage of atmospheric greenhouse gas carbon dioxide by enhanced biological processes. This can be done via increased rates of photosynthesis by via reforestation, sustainable forest management, and genetic engineering. Manipulation of these processes can enhance sequestration.

Peat Production

Peat bogs act as a sink for carbon because they accumulate partially decayed biomass that would otherwise continue to decay completely. By creating new bogs, or enhancing existing ones, the amount of carbon that is sequestered by bogs would increase.

Peat bogs

Forestry Practices

Forestry practices such as afforestation, proforestation, reforestation and urban forestry can be utilized to enhance carbon sequestration. Reforestation is the process of planting trees in a forest where the number of trees has been decreasing. Afforestation is when new trees are planted where there were no trees before, creating a new forest. Proforestation is the practice of protecting an existing forest, and allowing it to grow to its full potential so as to enhance carbon accumulation and structural complexity. Urban forestry is the care and management of tree populations in urban areas so as to enhance carbon sequestration over the trees' lifetime.

Wetlands

Wetland soil is found in coastal wetlands such as mangroves, sea grasses, and salt marshes. Wetlands are an important carbon sink, containing 20-30% of the world's soil carbon.

Agriculture

Cropland soils contain less soil organic carbon (SOC) as compared to natural vegetation by ~30-40%. This loss is due to the removal of plant material containing carbon, in terms of harvests. Agricultural practices can be modified by the following methods such as:

- Leaving harvest residues on field
- Perennial crop rotation, thus increasing the amounts of soc.

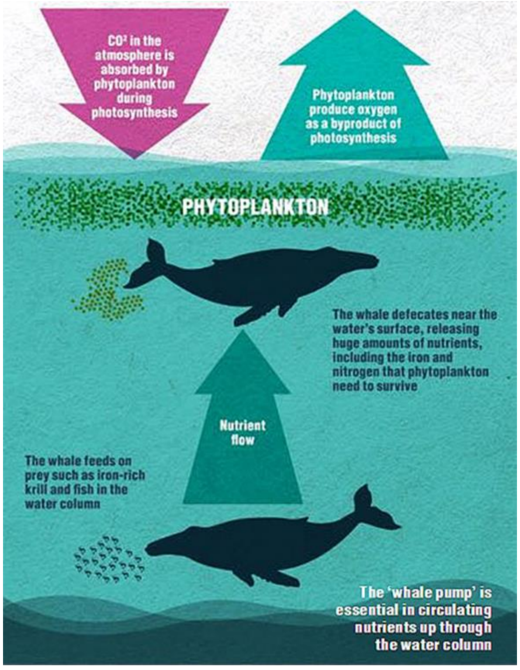
Deep Soil

Soils hold four times the amount of carbon stored in the atmosphere. About half of this is found deep within soils, stabilized by mineral-organic associations.

Reducing emissions: Increasing yields and efficiency generally reduces emissions as well, since more food results from the same or less effort. Techniques include more accurate use of fertilizers. E.g.: no-till farming requires less machine use and burns less fuel per acre.

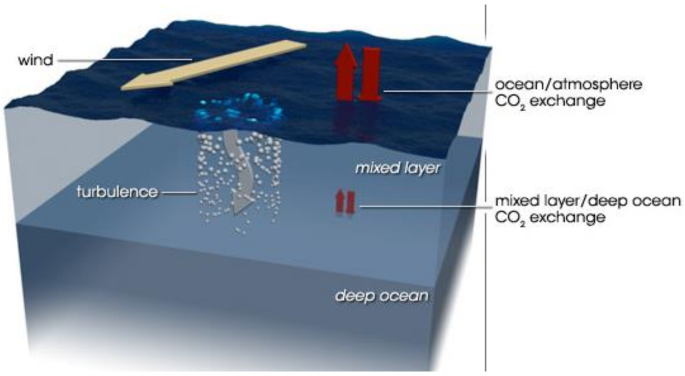
Oceanic Sequestration

Ocean iron and urea fertilization: It is a geoengineering technique. fertilization of ocean with urea and/or iron encourages phytoplankton growth, which removes carbon from the atmosphere. This technique is controversial as it not completely understood, and can result in release of nitrogen oxides that can disrupt the ocean's nutrient balance. But this process occurs naturally, mediated by sperm whales. Sperm whales act as agents of iron fertilization when they transport iron from the deep ocean to the surface during prey consumption and defecation. The iron-rich faeces cause phytoplankton to grow and take up more carbon from the atmosphere. When the phytoplankton dies, some of it sinks to the deep ocean and takes the atmospheric carbon with it.



Sperm whales defecate releasing large amounts of nutrients, which offset carbon

Mixing Layers



Encouraging various ocean layers to mix can move nutrients and dissolved gases around. Mixing can be achieved by placing large vertical pipes in the oceans to pump nutrient rich water to the surface, triggering blooms of algae, which store carbon when they grow and export carbon when they die. This produces results somewhat similar to iron fertilization.

Seaweed Farming

Seaweed grows in shallow and coastal areas, and capture significant amounts of carbon. Seaweed also grows fast, and can also be used to generate biomethane to produce electricity. It has been estimated that if seaweed farms covered 9% of the ocean, they could produce enough biomethane to supply Earth's equivalent demand for fossil fuel energy, remove 53 gigatonnes of CO2 per year from the atmosphere and sustainably produce 200 kg per year of fish, per person, for 10 billion people.

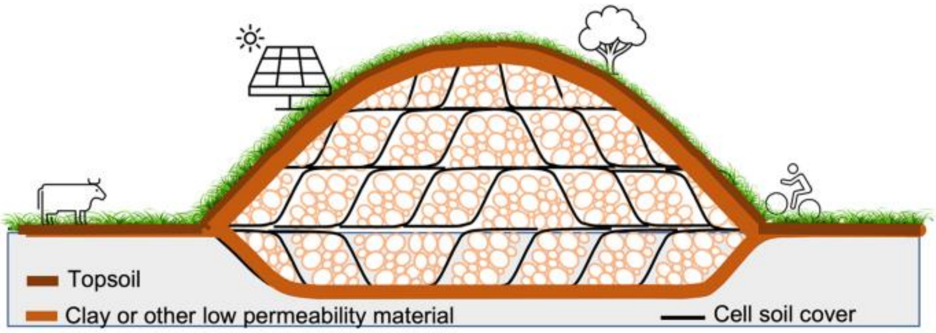

Underwater Seaweed Farming

Physical Sequestration

Physical sequestration methods make use of biomass burial, or use of biomass to capture carbon in different forms so as to create a terrestrial carbon sink. Let us review a few of these processes briefly:

Bioenergy with carbon capture and storage (BECCS)

BECCS is the process of extracting bioenergy from biomass, and capturing and storing the carbon, thereby removing it from the atmosphere. The carbon in the biomass comes from the greenhouse gas carbon dioxide (CO_2) which is extracted from the atmosphere by the biomass when it grows. Energy is extracted in useful forms (electricity, heat, biofuels, etc.) as the biomass is utilized through combustion, fermentation, pyrolysis or other conversion methods.



Wood Vault Version 1.1: Burial Mound (Tumulus)

Physical Burial

Burial of biomass (e.g., trees) directly, mimics natural processes that created fossil fuels.

Biochar Burial

Biochar is charcoal created by the pyrolysis of biomass waste. The resulting material is added to a landfill or used as a soil improver to create terra preta, in other words, fertile black soil.

Physical Oceanic Carbon Sequestration

Solubility of CO_2 is directly proportional to water pressure, and inversely proportional to temperature. This makes ocean beds, with high pressure and low temperature, the perfect long-term carbon sink. If CO_2 were to be injected to the ocean bottom, the pressures would be great enough for CO_2 to be in its liquid phase. This could create stationary pools of CO_2 at the ocean floor. The ocean could potentially hold over a thousand billion tons of CO_2. Let us review some of the commonly used methods for oceanic physical sequestration.

Dilute CO_2 injection

The carbon dioxide is usually injected at 1000 m depth to reduce carbon dioxide bubbles from escaping. These bubbles are dispersed up the water column via oceanic currents.

Release Of Solid Carbon Dioxide at Depth

CO_2 storage is facilitated via solid CO_2 or hydrate of CO_2 is 1.5 times heavier than seawater, thus it sinks to the ocean floor. Hydrate formation takes place when the dissolved concentration of Liquid Carbon Dioxide Is Around 30% And 400 Meters Below Sea Level.

Mineralization and deep-sea sediments

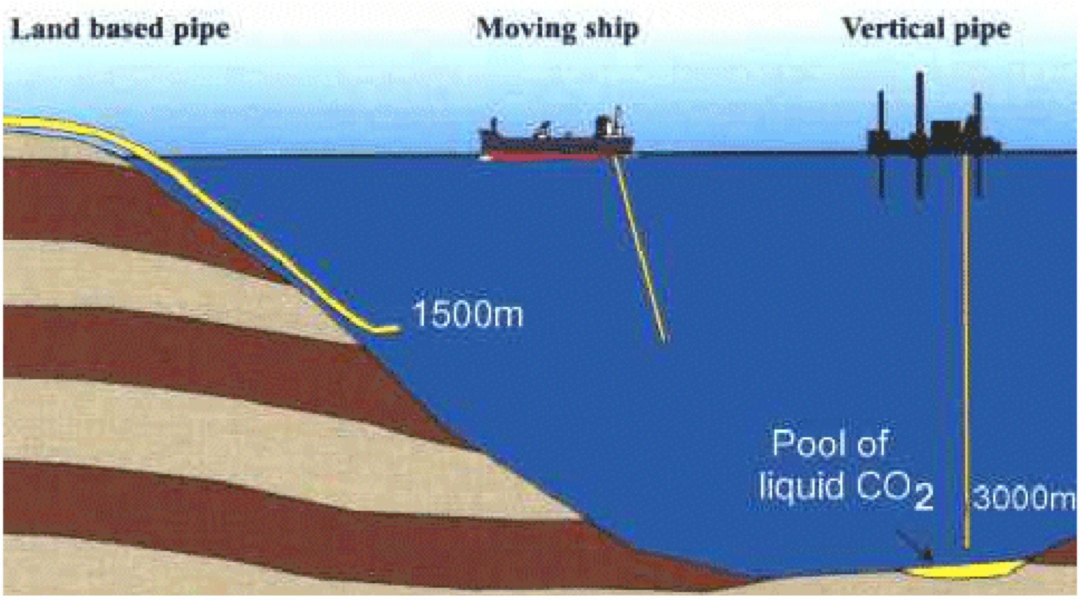
Similar to terrestrial processes, CO_2 mineralization can also occur under the sea. In addition, deep-sea sediment injects liquid CO_2 at least 3000 m below the surface, directly into ocean sediments to generate CO2 hydrates.

CO_2 Plumes

a mixture of dense CO_2 plumes and seawater is injected at 3 km depths, that sink due to their density, and are circulated by ocean currents.

CO_2 Lakes

Carbon dioxide lakes form on ocean floors in depressions or trenches through isolation. They also do not mix easily with the surface ocean since deep ocean has a very slow rate of mixing.



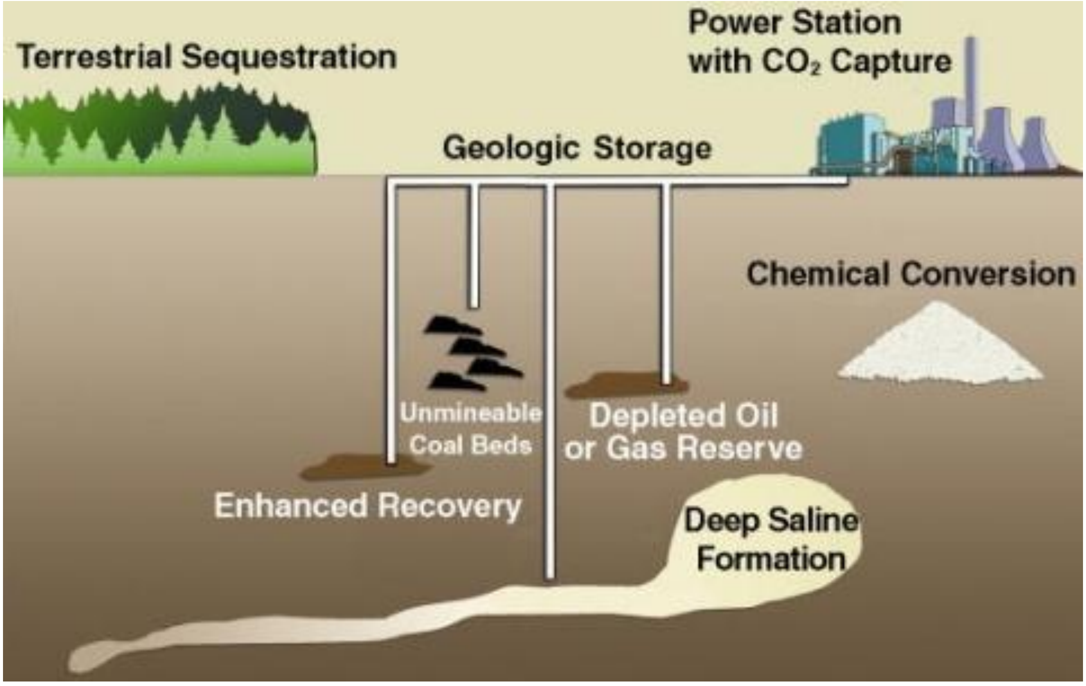
Environmental Impacts of Deep-Sea Ocean Sequestration

Deep-sea ocean sequestration is still largely unexplored. Scientists are studying their impacts through small-scale experiments. The spatial range of the ocean makes it extremely challenging to extrapolate results.

- Ocean sequestration in deep sea sediments has the potential to impact deep sea life. The chemical and physical composition of the deep sea does not undergo changes in the way that surface waters do. Due to its limited contact with the atmosphere, most organisms have evolved with very little physical and chemical disturbance and exposed to minimal levels of carbon dioxide. Deep sea ecosystems do not have rapid reproduction rates nor give birth to many offspring because of their limited access to oxygen and nutrients. Introducing lethal amounts of carbon dioxide into such an environment can have a serious impact on the population size and will take longer to recover relative to surface water species.
- Effects of pH vs CO_2: increased amounts of CO_2 cause acidification of water. Organisms are affected, not just by the acidification of water; but CO_2 itself interferes with their physiological function.
- Long-term effects: these are difficult to predict, but also important to understand, as it would impact not just deep oceans, but surface waters eventually as well

Physical Geo-sequestration

Geological sequestration refers to the storage of CO_2 underground in depleted oil and gas reservoirs, saline formations, or deep, un-minable coal beds. CO_2 released from fossil fuel combustion can be captured and compressed into a supercritical fluid, and injected deep underground, about 1 km depths, where it would be stable for up to millions of years.



Mineral carbonation

in this method, CO_2 is converted into stable carbonates of calcium or magnesium. removal and storage of CO_2 as calcium or magnesium carbonates; this reaction occurs naturally through the weathering of rocks over geologic time periods.